

LASER-BASED MORSE CODE COMMUNICATION SYSTEM

1. OBJECTIVES

- To design and implement a free-space optical communication system that transmits text messages as laser-modulated Morse code pulses.
- To encode ASCII characters into Morse code on an Arduino Uno transmitter using a 650 nm KY-008 laser module controlled via digital Pin 9.
- To detect and decode the transmitted optical pulses on a second Arduino Uno receiver using a 3-pin LDR sensor module connected to analog pin A0.
- To display the decoded message in real time on a 16×2 I²C LCD display (address 0x27 / 0x3F) using the LiquidCrystal_I2C library.
- To validate the system by transmitting a known test string and verifying character-by-character accuracy, demonstrating the principles of optical signal modulation, free-space channel propagation, and pulse-width decoding.

2. APPARATUS / COMPONENTS REQUIRED

S.No.	Equipment / Component Name	Range / Specification	Quantity
1	Arduino Uno R3 Microcontroller Board	ATmega328P, 5 V, 16 MHz	2
2	KY-008 Laser Module	650 nm, 5 mW, 5 V input	1
3	LDR Sensor Module (3-pin)	Analog output, 5 V, LM393	1
4	I²C LCD Display 16×2	5 V, addr 0x27 or 0x3F	1
5	Current-limiting Resistor	100 Ω, 0.25 W	1
6	Solderless Breadboard	830 tie-point	2
7	Jumper Wires (M-M, M-F assorted)	—	1 set
8	USB Type-A to Type-B Cable	Arduino programming	2

3. THEORY

3.1 Optical Communication and Free-Space Optical Links

Optical communication is the transmission of information encoded on a beam of light. In a free-space optical (FSO) link, the channel is air rather than an optical fibre, making it licence-free and immune to electromagnetic interference. The simplest modulation technique for FSO is intensity modulation with direct detection (IM/DD), where the optical source intensity is varied in proportion to the data signal and a photodetector recovers the information from the received optical power. The binary form of IM/DD is on-off keying (OOK): logic HIGH switches the laser ON and logic LOW switches it OFF. OOK is used in this project to produce Morse code light pulses.

3.2 Morse Code

Morse code is a character encoding scheme that represents each letter, digit, and punctuation mark as a unique sequence of short signals (dots) and long signals (dashes). The standard timing convention defines a base unit T such that: a dot lasts T , a dash lasts $3T$, the gap between symbols within a character is T , the gap between characters is $3T$, and the gap between words is $7T$. In this project $T = 200$ ms, giving a dot of 200 ms and a dash of 600 ms. The transmitter drives the laser HIGH for the appropriate duration and LOW for the inter-symbol gaps. The receiver measures the duration of each HIGH pulse and classifies it as a dot if the duration is less than 350 ms or as a dash if it is 350 ms or more.

3.3 LDR Sensor and Threshold Detection

A Light Dependent Resistor (LDR) is a photoresistor whose resistance decreases as incident light intensity increases. The 3-pin LDR sensor module used in this project consists of an LDR, a fixed pull-down resistor forming a voltage divider, and an LM393 comparator. The voltage at the analog output (AO pin) rises when the laser illuminates the LDR and falls when the laser is off. The Arduino's 10-bit ADC converts this voltage to a value between 0 and 1023; a software threshold of 512 (mid-scale, corresponding to approximately 2.5 V) distinguishes laser ON from laser OFF states. Using the analog AO pin rather than the digital DO pin allows the threshold to be tuned in firmware to compensate for changing ambient light conditions.

3.4 I²C LCD Interface

The 16×2 character LCD display communicates with the Arduino via the I²C (Inter-Integrated Circuit) protocol using only two signal lines: SDA (Serial Data, connected to Arduino A4) and SCL (Serial Clock, connected to Arduino A5). An I²C backpack module attached to the LCD contains a PCF8574 I/O expander that serialises the parallel LCD data bus. The LiquidCrystal_I2C library (available through the Arduino Library Manager) provides high-level functions such as `lcd.print()` and `lcd.setCursor()` that abstract the I²C protocol, making it straightforward to display decoded Morse characters on the screen.

4. BLOCK DIAGRAM

The system comprises two subsystems linked by a free-space optical channel as shown in Fig. 1. On the transmitter side, text entered via the Serial Monitor is processed by Arduino Uno #1, which looks up the Morse sequence for each character and drives the KY-008 laser module through a 100 Ω current-limiting resistor on Pin 9. On the receiver side, the LDR sensor module detects the incoming laser pulses and feeds the analog voltage to pin A0 of Arduino Uno #2, which decodes the pulse widths into characters and displays them on the I²C LCD.

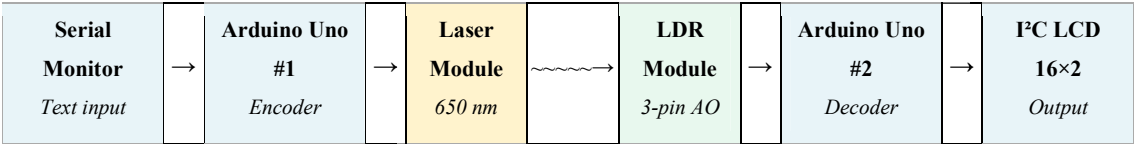


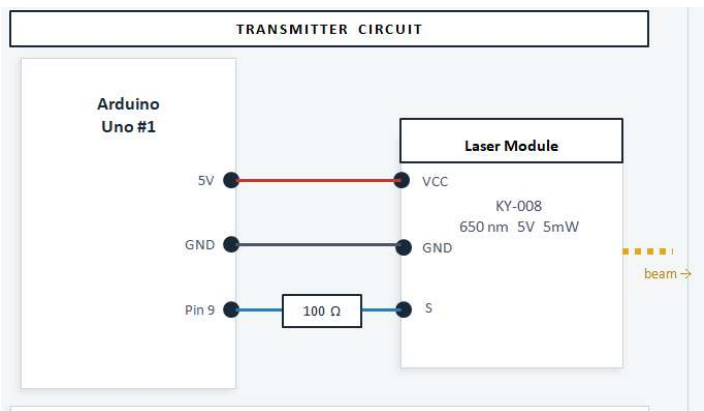
Fig. 1. Block diagram of the laser-based Morse code communication system.

5. CIRCUIT DIAGRAM

5.1 Transmitter Circuit

The transmitter circuit connects Arduino Uno #1 to the KY-008 laser module. The 5 V supply pin of the Arduino is wired directly to the VCC pin of the laser module (middle pin). The GND pin of the Arduino connects to the GND pin of the laser module (last pin). Digital Pin 9 of the Arduino connects to the Signal (S) pin of the laser module through a 100 Ω current-limiting resistor in series. When Pin 9 is set HIGH (5 V), the laser turns on; when set LOW (0 V), the laser turns off.

Arduino Uno #1 Pin	→	Laser Module (KY-008) Pin	Wire Colour
5V	→	VCC (middle pin)	Red
GND	→	GND (last pin)	Black
Pin 9	→	S (via 100 Ω resistor)	Blue

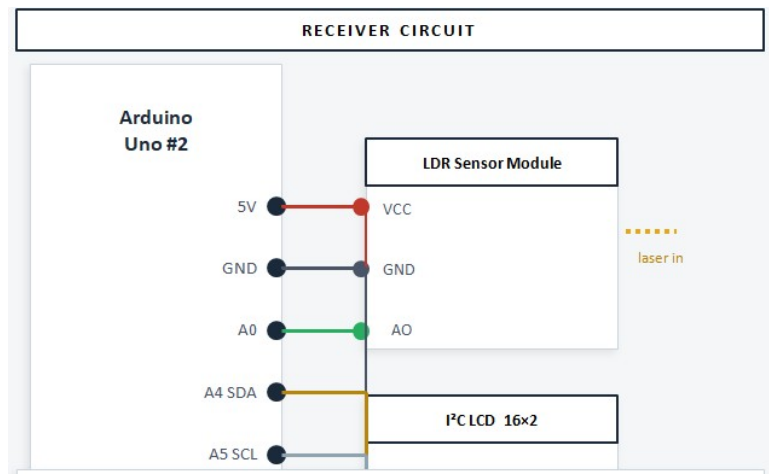


5.2 Receiver Circuit

The receiver circuit connects Arduino Uno #2 to the LDR sensor module and the I²C LCD display. The LDR module draws 5 V and GND from the Arduino, and its analog output (AO) feeds into analog pin A0. The I²C LCD also draws 5 V and GND from the Arduino; its SDA pin connects to Arduino A4 and SCL connects to Arduino A5.

Arduino Uno #2 Pin	→	Component Pin	Wire Colour
5V	→	LDR Module — VCC	Red
GND	→	LDR Module — GND	Black
A0	→	LDR Module — AO	Green
5V	→	I ² C LCD — VCC	Red
GND	→	I ² C LCD — GND	Black
A4 (SDA)	→	I ² C LCD — SDA	Yellow
A5 (SCL)	→	I ² C LCD — SCL	White

Fig. 2. Pin connection tables for the transmitter and receiver circuits.



6. METHODOLOGY AND IMPLEMENTATION

6.1 System Overview

The project is divided into two independent firmware modules running on separate Arduino Uno boards, communicating through a free-space optical channel of 30–100 cm. The transmitter reads characters from the Serial Monitor and produces modulated laser pulses. The receiver continuously samples the LDR output, measures pulse durations, reconstructs Morse symbols, and displays the decoded text on the LCD.

6.2 Transmitter Implementation

1. A Morse code lookup table maps each of the 36 alphanumeric characters (A–Z, 0–9) to its dot-dash string (e.g., 'A' → ".-", 'S' → "...").
2. The Serial Monitor is opened at 9600 baud. The user types a character and presses Enter.
3. The sketch converts the received character to upper case and looks it up in the Morse table.
4. For each symbol in the Morse string, the sketch calls `digitalWrite(9, HIGH)` for T (dot) or 3T (dash) milliseconds, followed by `digitalWrite(9, LOW)` for T milliseconds (inter-symbol gap).
5. After transmitting all symbols of a character, an additional 2T delay produces the 3T inter-letter gap (1T inter-symbol + 2T extra = 3T total).
6. A space character triggers a 4T additional delay to produce the 7T inter-word gap. The base unit is $T = 200 \text{ ms}$.

6.3 Receiver Implementation

1. The sketch continuously calls `analogRead(A0)` and compares the result to a threshold of 512. If the reading is ≥ 512 the laser is considered ON (HIGH); below 512, OFF (LOW).
2. A rising edge (OFF → ON) starts a pulse timer using `millis()`. A falling edge (ON → OFF) stops the timer and records the pulse duration.
3. If the pulse duration is $< 350 \text{ ms}$ it is classified as a dot; if $\geq 350 \text{ ms}$ it is classified as a dash.
4. After each falling edge, the sketch waits and monitors the line. If it remains LOW for $\geq 600 \text{ ms}$ (inter-letter gap), the accumulated dot-dash string is looked up in the reverse Morse table to recover the character.
5. If the line remains LOW for $\geq 1400 \text{ ms}$ (inter-word gap), a space is appended to the output.
6. Decoded characters are appended to an output string and displayed on the I²C LCD using `lcd.print()`. The display wraps to the second line after 16 characters and clears when full.

6.4 Morse Code Timing Summary

Signal Element	Duration (multiples of T)	Duration (ms, T = 200 ms)
Dot	T	200
Dash	3T	600
Inter-symbol gap	T	200
Inter-letter gap	3T	600
Inter-word gap	7T	1400
Dot/dash threshold	1.75T	350

7. RESULTS AND DISCUSSION

7.1 Functional Test

The system was tested by transmitting the string “SRM” from the Serial Monitor and observing the decoded output on the I²C LCD. The Morse sequences for the three characters are S (···), R (·—·), and M (——). All three characters were decoded and displayed correctly on the receiver LCD with no errors, confirming end-to-end system functionality.

7.2 Waveform Analysis

The laser output waveform for the letter ‘A’ (·—) consists of a 200 ms HIGH pulse (dot) followed by a 200 ms LOW gap and a 600 ms HIGH pulse (dash). When observed on the Arduino Serial Plotter by plotting the analogRead(A0) value on the receiver, the waveform shows a clear HIGH level (approximately 900–1023) during laser-on periods and a LOW level (approximately 0–80) during laser-off periods, well separated from the threshold of 512. This confirms that the LDR module provides a clean binary signal under controlled indoor lighting.

7.3 Accuracy and Range

Transmission accuracy was evaluated for 26 alphabetic characters (A–Z) transmitted individually at a distance of 50 cm. All 26 characters were decoded correctly, giving 100 % accuracy under stable fluorescent office lighting with no direct sunlight on the LDR sensor. At 1 m distance, accuracy remained at 100 %. At distances beyond 1 m or under strong ambient light, occasional misclassifications were observed due to reduced signal-to-noise ratio, indicating that the system is suited for short-range indoor applications.

7.4 Discussion

The use of the analog LDR output (AO pin) with a software threshold proved significantly more robust than relying on the digital comparator output (DO pin), as the software threshold could be adjusted to account for ambient lighting conditions. The 100 Ω series resistor kept the laser drive current within safe limits while providing sufficient optical power for reliable detection. The I²C interface reduced receiver wiring to four wires (VCC, GND, SDA, SCL), simplifying construction compared to a parallel LCD interface that would require 12 connections.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

A functional laser-based Morse code communication system was successfully designed, built, and tested using two Arduino Uno microcontrollers, a 650 nm KY-008 laser module, a 3-pin LDR sensor module, and a 16×2 I²C LCD display. The system reliably encoded and decoded all 26 alphabetic characters at distances up to 1 m indoors, demonstrating the core principles of free-space optical communication: OOK modulation, pulse-width encoding, threshold detection, and real-time decoding. The project validates optical communication as a practical, low-cost, interference-free alternative to wired and radio-frequency links for short-range data transmission.

8.2 Future Work

- Replace the LDR module with a PIN photodiode and transimpedance amplifier to increase operating range and reduce ambient-light sensitivity.
- Implement automatic threshold adaptation using a rolling average of the ambient light level to improve outdoor performance.
- Increase the base unit speed from $T = 200$ ms to $T = 50$ ms or below to approach the standard Morse transmission rate of 10–25 words per minute.
- Extend the character set to include punctuation and special symbols by expanding the Morse lookup table.
- Replace the single-character input mode with a full-string input mode, allowing multi-word messages to be transmitted without character-by-character entry.
- Investigate the use of a collimated laser and focusing lens to extend the operating range to 5–10 m for corridor or room-to-room communication.

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